

Privacy Challenges in Healthcare AI: The DeepMind Case

Abstract

On the one hand, the integration of artificial intelligence into healthcare has opened significant possibilities for improving medical diagnosis and patient care. On the other hand, it has also introduced serious challenges in terms of patient privacy, informed consent, and the accountability of those who collect and use health data. These challenges have become particularly visible in collaborations between public health institutions and private technology companies, resulting in partnerships that are increasingly common; however, insufficiently regulated.

This paper examines the 2015–2017 data-sharing agreement between Google DeepMind and the Royal Free NHS Trust. A case study observing how such partnerships can sometimes fail patients, raising privacy and ethical concerns. Under this agreement, the medical records of approximately 1.6 million patients, originally intended for developing Streams clinical application, were transferred to DeepMind without an explicit consent. The case covers three interconnected problems that are still relevant. Namely, it analyses the absence of meaningful informed consent, the secondary use of data collected for an entirely different purpose, and a lack of transparency.

The analysis is conducted within two key regulatory frameworks with ethical matters in mind. Under the General Data Protection Regulation, the agreement constitutes a clear violation of the principles of purpose limitation, data minimization, and the right to an explanation of automated decisions. Under the EU AI Act, the first binding legal framework specifically addressing artificial intelligence, which entered into force in August 2024, healthcare AI systems are classified as high-risk, requiring mandatory Data Protection Impact Assessments, human oversight mechanisms, and transparency obligations. However, none of these safeguards were met in the DeepMind case, further offering a new perspective on the above mentioned case regarding the patient data privacy.

Finally, the paper considers federated learning and differential privacy as concrete technical approaches that could enable future healthcare AI development without centralizing sensitive patient data. The paper concludes that the DeepMind case remains a relevant matter for any future collaboration between public

healthcare systems and technology companies, and that the regulatory tools to prevent such situations now exist, what remains is the institutional will to apply them consistently.

Keywords: EU AI Act; informed consent; secondary use of data; federated learning; differential privacy

References

- [1] K. Moulaei et al., “Patient consent for the secondary use of health data in artificial intelligence (AI) models: A scoping review,” *International Journal of Medical Informatics*, 2025. DOI: <https://pubmed.ncbi.nlm.nih.gov/40107041/>
- [2] D. A. Tejada-Arango et al., “Shaping the future of AI in healthcare through ethics and governance,” *Humanities and Social Sciences Communications*, 2024. Available at: <https://www.nature.com/articles/s41599-024-02894-w>
- [3] European Parliament and Council of the European Union, “Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act),” *Official Journal of the European Union*, OJ L 2024/1689, 12.7.2024. Available at: <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>
- [4] A. Benis et al., “Medicine, healthcare and the AI Act: gaps, challenges and future implications,” *European Heart Journal – Digital Health*, 2025. DOI: <https://academic.oup.com/ehjdh/article/6/4/833/8118685>
- [5] B. A. Mir, S. R. Abbas, and S. W. Lee, “Federated Learning in Healthcare Ethics: A Systematic Review of Privacy-Preserving and Equitable Medical AI,” *Healthcare (MDPI)*, vol. 14, no. 3, p. 306, 2026. DOI: <https://doi.org/10.3390/healthcare14030306>
- [6] A. Ziller et al., “Differential privacy for medical deep learning: methods, tradeoffs, and deployment implications,” *npj Digital Medicine*, 2026. DOI: <https://www.nature.com/articles/s41746-025-02280-z>